Size Confounds Semantics: An Audit of SING's Null-Space Leakage Score

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Abstract

SING [10] ranks classifiers by how much class-relevant semantics leak into the linear head's null space, and reports that a "DinoViT" pretrained with DINO leaks least. We audit that claim against public checkpoints and against size controls. No public ViT-L/16 that matches SING's code or wrappers is a DINO model: every candidate is supervised or SWAG-supervised, and none reproduces the reported 84.0% top-1. Inside a single model, the absolute-shift score |AS| itself grows by a factor of $\approx$10 when the removed null subspace grows from d=24 to d=1048 (3 of 4 models). At matched d=24 the ranking already disagrees with SING's published order (Kendall τ=0.29). Fitting |AS| against removed feature energy ρ across pooled cells gives $|\mathrm{AS}| \propto \rho^{0.43}$, near the $\sqrt{\rho}$ geometric prediction, but the energy-vs-dimension separation is mixed rather than decisive. Dividing null removal by an energy-matched random-subspace baseline defines a size-fair score $\mathrm{AS}_{\mathrm{fair}}$. On the 13 SING models, $\mathrm{AS}_{\mathrm{SING}}$ recovers the published Fig. 6b order (τ=0.82) while $\mathrm{AS}_{\mathrm{fair}}$ does not (τ=$-$0.18); two class-level claims reverse under bootstrap CIs. Neither score recovers planted leakage well. We recommend reporting ρ and $\mathrm{AS}_{\mathrm{fair}}$ next to raw AS whenever null-space scores are used to compare architectures.

1 Introduction

A linear classifier on features $f \in \mathbb{R}^m$ has a null space of dimension $m-c$ (for c classes) that leaves the logits unchanged. SING [10] calls class-relevant content in that null space "leakage": material the head could have used but did not. Absolute Shift (AS) measures how far a CLIP-aligned [5] translation of f moves once the null component is stripped out. SING ranks 13 ImageNet models by AS (and by the IS/AS ratio) and singles out a ViT labeled "DinoViT" as the least leaky [1].

Related null-space tools appear in OOD detection [2, 9] and overfitting scores [6]. Across architectures the null dimension d=$m-\mathrm{rank}(W)$ and the null energy fraction $\rho = \mathbb{E}\|V_n V_n^\top f\|^2 / \mathbb{E}\|f\|^2$ both swing by orders of magnitude. That raises a basic confound: if AS largely tracks how much energy is removed, then "semantics" rankings across models are partly rankings of null size.

Contributions. (1) SING's "DinoViT" is not identifiable as DINO—every matching public candidate is supervised, and none reproduces 84.0% top-1. (2) Within-model AS(d) curves give a $\approx 10\times$ rise from d=24 to d=1048 on ResNet50, ResNeXt101 and DINOv2-L; BiT [3] is the exception, explained by translator cancellation. (3) Energy and dimension are hard to separate: a pooled fit yields exponent a=0.43 on ρ against a $\sqrt{\rho}$ prediction of 0.5, yet fixed-ρ flatness is not clean. (4) On 13 models, size-fair $\mathrm{AS}_{\mathrm{fair}}$ breaks SING's published ranking (τ: 0.82 $\rightarrow$ $-$0.18) and reverses several claims, with the caveats below. $\mathrm{AS}_{\mathrm{fair}}$ is offered as a reporting correction, not as a new leakage oracle.

2 Background and the DinoViT identity

SING AS in brief. Let $T(f) = Af + b$ be SING's released linear translator into CLIP space and V_n an orthonormal basis of $\ker(W)$ (tolerance 10^{-5}). For a text direction t, SING's per-image absolute shift is the change in angle $\theta(T(f), t)$ after removing $V_n V_n^\top f$. $\mathrm{AS}_{\mathrm{SING}}$ denotes the image-averaged |AS| under full null removal (class text unless noted).

Exp 0b: "DinoViT" is not DINO. SING reports DinoViT at 84.0% ImageNet top-1 (Supp. Table 2). The released models.yaml loads a timm AugReg ViT-L/16; the supplement's wrapper uses torchvision ViT attribute names. We ran five public ViT-L/16 candidates on the full 50k validation set (native and force-224): torchvision IMAGENET1K_V1 (79.67%), timm AugReg (85.85%), SWAG linear [7] (85.14%), SWAG E2E (88.07%), and HF google/vit-large-patch16-224 (81.98%). None lands in 84.0 ± 0.3 (Fig. 1). All five are supervised or SWAG weakly supervised. DINO never shipped a ViT-L with a 1000-way head [1]. The DINO attribution in the headline result therefore has no identifiable public checkpoint behind it.

Exp 0 (re-analysis). Digitizing SING Supp. Fig. 6b IS/AS bars and correlating against measured null dimension gives Spearman -0.46 (p=0.12) over 13 models (DenseNet null dim = 27 at tol 10^{-5}). Null energy is far more predictive: $\mathrm{Spearman}(\rho_{\mathrm{full}}, \mathrm{IS/AS}) = -0.94$ (p=1.9×10^{-6}). DeiT3-L [8] has null dim 24 yet the worst

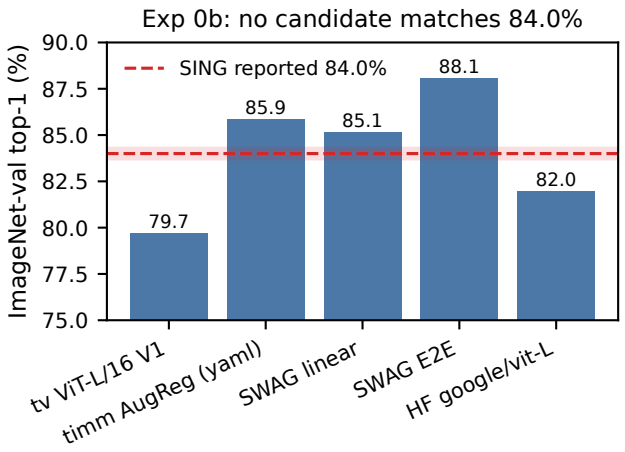


Figure 1: Exp 0b top-1 on ImageNet-val. No candidate matches SING's reported 84.0%.

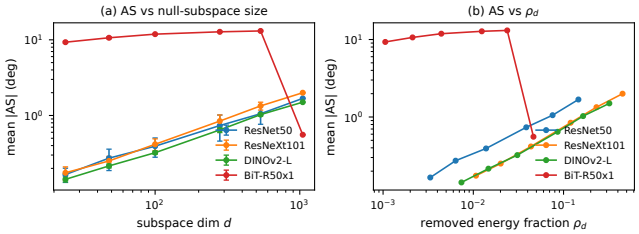


Figure 2: Exp 1: mean |AS| vs. subspace size d (a) and removed energy ρ_d (b). BiT collapses at the full null (translator cancellation).

ratio, so dimension alone cannot explain the ranking. That is the motivation for a within-model test.

3 Within-model size dependence (Exp 1)

We fix a 10k ImageNet-val subset (10 images/class) and remove random d-dimensional subspaces of each model's own null space ($d \in \{24, 48, 100, 280, 536, 1048\}$, 20 seeds for $d<d_{\text{null}}$) for ResNet50, ResNeXt101, BiT-R50x1 and DINOv2-L (all d_{null}=1048), using SING's released translators.

Fig. 2 and Table 1 show r=AS(1048)/AS(24) $\approx$ 10 on 3 of 4 models. Matching d=24 over 8 models yields Kendall τ=0.29 (p=0.40) vs. the Fig. 6b order for |AS|. Both pre-registered GO conditions hold.

BiT anomaly. BiT's full-null AS collapses (r=0.06) even though partial subspaces give large AS (∼9–13°). The singular values of AV_n are extreme (top 284 vs. 5.6 for ResNet50): orthogonal null blocks nearly cancel through the translator when summed, so full-null AS understates BiT's per-direction sensitivity. Translator conditioning, not a code bug.

Model	r	2σ band	R^2 vs d	R^2 vs ρ_d
ResNet50	10.15	[5.83, 14.48]	0.983	0.983
ResNeXt101	11.44	[6.93, 15.95]	0.977	0.977
DINOv2-L	10.48	[8.85, 12.10]	0.974	0.974
BiT-R50x1	0.060	[0.057, 0.063]	0.512	0.512

Table 1: Exp 1: r=AS(1048)/AS(24). Energy and dimension fits are collinear for random subspaces.

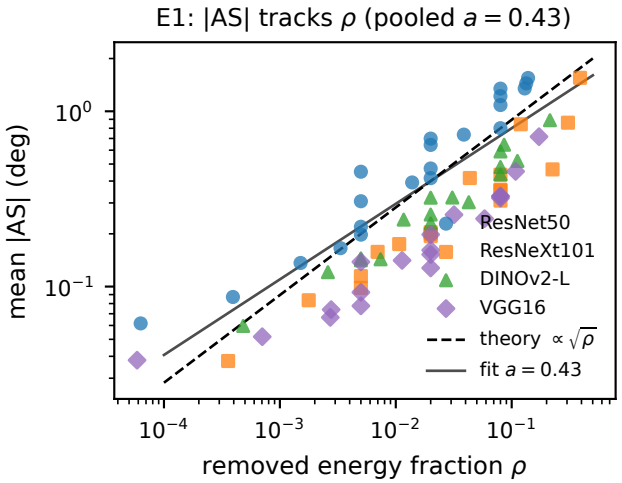


Figure 3: E1 cells: |AS| vs. ρ. Dashed: $\sqrt{\rho}$ theory; solid: fitted a=0.43.

Caveat. For random subspaces ρ_d is nearly linear in d, so R^2 vs. d and vs. ρ_d coincide ($\approx$ 0.98). Exp 1 alone cannot separate energy from dimension.

4 Energy vs. dimension (E1): a mixed result

Geometry. Suppose a linear translator maps a removed feature component of relative energy ρ into CLIP space. To first order the angular displacement scales with the removed magnitude $\|\Pi f\| \propto \sqrt{\rho}$ when the residual is small relative to $\|T(f)\|$. So $|\text{AS}| \propto \rho^{1/2}$ at fixed geometry—an exponent of 0.5.

Design. On ResNet50, ResNeXt101, DINOv2-L, VGG16 and EVA02-L we (a) compare high-energy vs. low-energy null PCs at fixed $d \in \{24, 100, 280\}$ and (b) hit fixed $\rho \in \{0.5\%, 2\%, 8\%\}$ at several d by mixing top/bottom PCs. Pooled fit over 76 feasible cells:

$$\log|\text{AS}| \;=\; a\log\rho \;+\; b\log d \;+\; c_{\text{model}},$$

gives a=0.431 [0.430, 0.433], b=0.010 [0.008, 0.012], R^2=0.93 (Fig. 3).

High/low energy ratios at fixed d exceed 3× on all 4 evaluable models (e.g. ResNet 11–22×, ResNeXt ≈ 10×, DINOv2 3.7–5.1×, VGG16 6–10×). Fixed-ρ max/min across d still exceeds 1.5 on several cells (ResNet up to 2.29 at ρ=0.5%). EVA02-L (d_{null}=24) is not evaluable

beyond its full-null cell. E1 is mixed, not decisive. Energy dominates the fit and comes within ≈ 0.07 of the $\sqrt{\rho}$ exponent, but the pre-registered ENERGY-DRIVEN flatness rule is not cleanly met.

5 A size-fair score

Take an energy-matched random baseline: draw a Haar subspace of the full feature space that removes exactly the same energy $\mathbb{E}\|\Pi f\|^2$ as the null projector, and let $\mathrm{AS}_{\mathrm{rand}}$ be its mean |AS| (20 draws). Then

$$\mathrm{AS}_{\mathrm{fair}} = \frac{\mathrm{AS}_{\mathrm{null}}}{\mathrm{AS}_{\mathrm{rand}}}.$$

Near 1, the null is no more shifting than a random equal-energy subspace; $\gg 1$ means concentrated null leakage. This is the E3-registered definition, reused at α=0 for the E2 audit.

6 Planted leakage (E3)

We plant a CLIP concept direction (red/striped/water) into each of 8 models' null spaces at log-uniform $\alpha \in [0.01, 1]$ over 50 trials, leaving logits unchanged (max $|\Delta\text{logit}| \leq 2.3\times10^{-5}$). Kendall τ vs. the planted-α ranking, concept-averaged: $\mathrm{AS}_{\mathrm{SING}}$ 0.386 $[0.324, 0.447]$; $\mathrm{AS}_{\mathrm{fair}}$ 0.071 $[0.004, 0.138]$. Neither recovers planted strength well; $\mathrm{AS}_{\mathrm{fair}}$ sits near chance. Per-concept means look similar (red 0.386/0.070, striped 0.403/0.061, water 0.369/0.083). A few quirks: DeiT3-L signed AS is negative at every α and |AS| shrinks as α grows; BiT $\mathrm{AS}_{\mathrm{SING}}$ reaches 17–20° at α=1 while its $\mathrm{AS}_{\mathrm{fair}}$ at α=0 is only 0.045–0.063; $\mathrm{AS}_{\mathrm{fair}}$ is non-monotonic in α for EVA02-L and ConvNeXt-B. Size-fairness and planted-ranking recovery pull in different directions. We propose $\mathrm{AS}_{\mathrm{fair}}$ only as a size correction for cross-model AS comparisons, not as a calibrated leakage oracle.

7 Audit of SING's 13-model claims (E2)

On the same 10k subset plus a 200-image claim-class set (all 50 val images of jellyfish, porcupine, Arabian camel, sports car), we recompute $\mathrm{AS}_{\mathrm{SING}}$ and $\mathrm{AS}_{\mathrm{fair}}$ for all 13 released translators.

Ranking. Against the digitized Fig. 6b IS/AS order: $\tau(\mathrm{AS}_{\mathrm{SING}})$=0.821 ($p$=1.3×10^{-5}); $\tau(\mathrm{AS}_{\mathrm{fair}})$=−0.179 ($p$=0.435). Dropping DenseNet: $\tau(\mathrm{AS}_{\mathrm{fair}})$=−0.364. Mean over 173 SING concepts: τ=0.641 vs. −0.026. $\mathrm{AS}_{\mathrm{SING}}$ recovers SING; the disagreement shows up only after the size correction (Figs. 4–5).

Claim reversals. Under $\mathrm{AS}_{\mathrm{fair}}$, claims C1–C7, C9, C11, C12 reverse relative to $\mathrm{AS}_{\mathrm{SING}}$ (DinoViT is no longer least leaky; ResNeXt is no longer among the worst). Two claims reverse under both scores, with bootstrap CIs:

- C8 (ResNet porcupine / sports-car ≥90th percentile): porcupine percentile 84.8 $[80.1, 87.7]$ ($\mathrm{AS}_{\mathrm{SING}}$), 68.4 $[64.5, 71.5]$ ($\mathrm{AS}_{\mathrm{fair}}$); sports car 61.0 $[56.1, 66.2]$ and 67.2 $[63.6, 71.4]$.
- C10 (DinoViT jellyfish ≫ camel): jellyfish−camel CI $[-0.205, -0.010]$ ($\mathrm{AS}_{\mathrm{SING}}$) and $[-1.403, -0.458]$ ($\mathrm{AS}_{\mathrm{fair}}$)—opposite in sign to SING's claim.

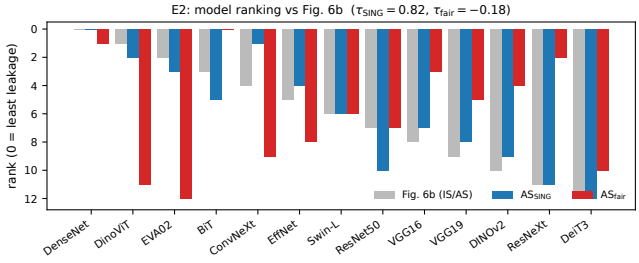


Figure 4: Ranks (0=least leakage). $\mathrm{AS}_{\mathrm{SING}}$ tracks Fig. 6b; $\mathrm{AS}_{\mathrm{fair}}$ does not.

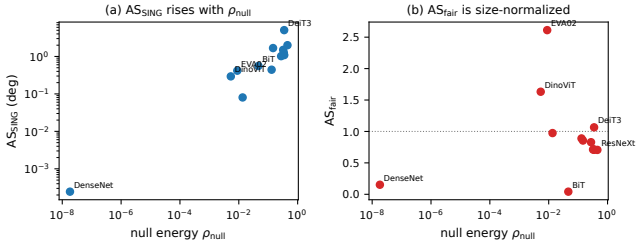


Figure 5: (a) $\mathrm{AS}_{\mathrm{SING}}$ rises with ρ_{null}. (b) $\mathrm{AS}_{\mathrm{fair}}$ removes that trend.

Caveats (must-state). (i) C1/C2 are IS/AS-ratio claims tested here on |AS|; C11 is a steering claim proxied by null-removal AS; C10 uses class-text AS on 50 images/class (SING used ≈30 concepts / ≈1300 images). (ii) DenseNet has ρ_{null}=1.8×10^{-8}, so $\mathrm{AS}_{\mathrm{fair}}$ is degenerate. (iii) C13 (EVA02 as target space) is untestable: no EVA02-space translators were released. (iv) Our DinoViT IS/|AS| is 20.95 vs. 70.16 in Fig. 6b—absolute ratios differ, but the $\mathrm{AS}_{\mathrm{SING}}$ ranking still agrees.

Under $\mathrm{AS}_{\mathrm{fair}}$, DinoViT sits near the worst end (12th of 13 by class $\mathrm{AS}_{\mathrm{fair}}$; EVA02 worst); under $\mathrm{AS}_{\mathrm{SING}}$ it ranks 3rd. BiT becomes best under $\mathrm{AS}_{\mathrm{fair}}$ (0.042) because random subspaces of matched energy already move AS by 13.2°—the same translator-cancellation regime as Exp 1. DenseNet's near-zero ρ makes $\mathrm{AS}_{\mathrm{fair}}$ numerically unstable, so we also report τ without it (−0.364).

8 Discussion

SING's within-model AS is a useful probe of translator-mediated null content. Used as a cross-model score, it conflates semantics with how large and energetic

ID	Claim (short)	AS_{SING}	AS_{fair}
C1	DinoViT best IS/AS among 5	holds	reverses
C2	ResNeXt lowest IS/AS among 5	holds	reverses
C3	ResNet leaks more than DinoViT	holds	reverses
C4	DinoViT least class leakage	holds	reverses
C5	ResNeXt high AS (top-2 of 5)	holds	reverses
C6	ResNet per-class AS > DinoViT	holds	reverses
C7	DinoViT per-class |AS|<1	holds	reverses
C8	Porcupine/sports-car ≥90th pct.	reverses	reverses
C9	No ResNet–DinoViT class corr.	holds	reverses
C10	Jellyfish ≫ camel (DinoViT)	reverses	reverses
C11	DinoViT least steerable (proxy)	holds	reverses
C12	Fig. 6b 13-model order	holds	reverses
C13	EVA02 target-space order	untestable	untestable

Table 2: E2 claim audit. C1/C2/C11 use |AS| proxies; C10 is class-text on 50 imgs/class.

the null is. A boring baseline already said as much: over 13 models, Spearman($\rho_{\text{full}}, |\text{AS}|_{\text{full}}$)=0.87 and Spearman($\rho_{\text{full}}, \text{IS/AS}$)=−0.94. On top of that, the DinoViT label has no public checkpoint behind it.

What AS_{fair} is and is not. It is a reporting correction: divide null AS by an energy-matched random baseline so models are not rewarded for a small or low-energy null. It is not a validated ground-truth leakage measure. E3 shows that neither score recovers planted α rankings at the pre-registered $\tau \geq 0.8$ bar, and AS_{fair} is worse at that particular task. We do not claim any replacement score beyond this size correction.

Recommendations. For any null-space leakage comparison:

1. Name the exact checkpoint and measured top-1 (avoid unverifiable labels such as "DinoViT").
2. Report d_{null} and ρ_{null} next to the score.
3. Report AS_{fair} (or another energy-matched baseline) beside raw AS.
4. Flag translator condition numbers when full-null and block-wise AS disagree (BiT).
5. Prefer within-model interventions (subspace size, planted signal) over cross-architecture league tables when d and ρ vary widely.

Limitations. E1 is mixed rather than decisive on energy vs. dimension. E3 shows AS_{fair} does not recover planted α. Several E2 claims use proxies (C1/C2/C10/C11); DenseNet's AS_{fair} is degenerate; C13 is untestable. ImageNet-derived features restrict redistribution. We did not retrain translators, evaluate a true DINO ViT-B/16 [1], or extend the size audit to ViM [9] / Cook [2] / Rezaei [6] scores [4].

9 Conclusion

SING's AS ranking of 13 classifiers reproduces with the released translators, but it is size-confounded. Correcting for null energy with AS_{fair} breaks the published order (τ: 0.82 → −0.18) and reverses multiple claims, and the "DinoViT" result cannot be pinned on DINO. Until a calibrated leakage oracle exists, cross-model null-space comparisons should report ρ and a size-fair baseline alongside raw AS.

Reproducibility. Analysis code and aggregated tables: https://github.com/sudeshgm/ic014-sing-null-dimension. ImageNet features are private.

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A Additional numerical detail

Exp 1 full-null anchors. Mean |AS| at full null: ResNet50 1.682°, ResNeXt101 1.999°, BiT 0.556°, DINOv2-L 1.499°; at d=24: 0.166, 0.175, 9.30, 0.143 (seed SD in RESULTS.md).

E2 per-model class scores. Table 3 lists ρ_{null}, AS_{SING}, AS_{rand} and AS_{fair} used for the ranking.

Model	ρ_{null}	AS_{SING}	AS_{rand}	AS_{fair}
VGG16	0.350	1.088	1.537	0.708
VGG19	0.339	1.256	1.755	0.716
DenseNet121	1.8e-8	0.00025	0.0016	0.152
ResNet50	0.146	1.676	1.960	0.855
DinoViT	0.0054	0.294	0.180	1.632
EffNet-B0	0.130	0.443	0.498	0.889
BiT-R50x1	0.046	0.557	13.24	0.042
ResNeXt101	0.451	1.999	2.829	0.707
ConvNeXt-B	0.0135	0.080	0.083	0.974
Swin-L	0.274	1.019	1.229	0.829
DINOv2-L	0.322	1.499	2.105	0.712
DeiT3-L	0.348	5.048	4.738	1.065
EVA02-L	0.0089	0.417	0.160	2.612

Table 3: E2 class-text scores on the 10k subset (from e2_models.csv).

E3 per-concept τ. AS_{SING} / AS_{fair}: red 0.386 / 0.070; striped 0.403 / 0.061; water 0.369 / 0.083.